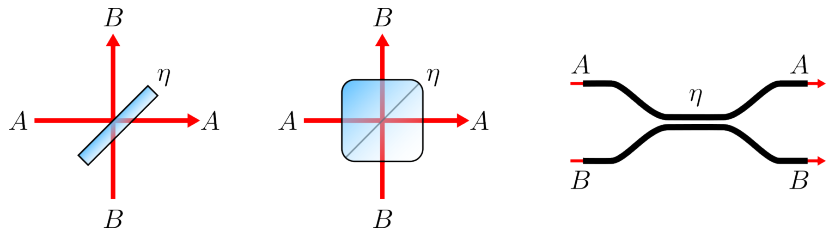

Figure 3.11: Different conventions for depicting beam splitters commonly seen in the scientific literature. *Left and middle:* free-space, bulk optics beam-splitters. *Right:* Fibre or wave-guide beam splitter.

they correspond to transformations of the annihilation operators of the form

$$\hat{a}_k \to U_{k1}\hat{a}_1 + U_{k1}\hat{a}_2 + \cdots + U_{k1}\hat{a}_N = \sum_{l=1}^{N} U_{kl}\hat{a}_l, \tag{3.133}$$

where the matrix $\boldsymbol{U}$ with elements U_{kl} must be unitary. Eqivalently, collecting the annihilation operators in a vector $\hat{\boldsymbol{a}}$ then $\hat{\boldsymbol{a}} \to \boldsymbol{U}\hat{\boldsymbol{a}}$. The unitarity of $\boldsymbol{U}$ ensures that the canonical commutation relations are preserved. Furthermore, because (3.133) does not mix annihilation and creation operators, energy is conserved.

In practice, there are a lot of different physical components available that all implement transformations of the form (3.133) (at least up to small imperfections such as loss) and the same transformation may be implemented in various different ways. Which implementation is possible or preferred will depend on the practical setup. However, formally every passive linear-optical transformation can be constructed from just two basic operations: phase shifts and beam splitters.

Phase shifts

Phase shifts have already been discussed in Sec. 3.1.4. They can be implemented by changing the optical path length of one mode relative to other modes in the system (only relative phases changes are observable). In free-space optics, this can be done by changing the spatial path length or by inserting an element with a different index of refraction, such as a piece of glass, into the path. In optical fibres, phase shifts can be achieved by applying tension, e.g. by slightly stretching the fibre.

Beam splitters

Beam splitters act on a pair of optical modes. They allow some of the light from each mode to be transmitted or reflected into the other, creating interference (see Fig. 3.11). For two modes A, B with annihilation operators $\hat{a}$ and $\hat{b}$, the

beam-splitter operator $\hat{U}_{BS}(\eta)$ transforms[6]

$$\hat{U}_{BS}(\eta)\,\hat{a}\,\hat{U}^{\dagger}_{BS}(\eta) = \sqrt{\eta}\,\hat{a} + \sqrt{1-\eta}\,\hat{b}, \tag{3.134}$$

$$\hat{U}_{BS}(\eta)\,\hat{b}\,\hat{U}^{\dagger}_{BS}(\eta) = \sqrt{1-\eta}\,\hat{a} - \sqrt{\eta}\,\hat{b}. \tag{3.135}$$

The parameter $0 \leq \eta \leq 1$ is the *transmittance*. It corresponds to the fraction of input light in each input mode which remains in the same mode. Similarly, the *reflectivity* $1-\eta$ is the fraction transferred into the other mode. We will sometimes drop the explicit dependence when η is clear from the context, writing just $\hat{U}_{BS}$ for $\hat{U}_{BS}(\eta)$.

As may be expected, applied to a state with no photons, a beam splitter does nothing

$$\hat{U}_{BS}(\eta)|vac\rangle = |vac\rangle, \tag{3.136}$$

where $|vac\rangle$ denotes the joint vacuum state of all modes. For an input state with a single photon in mode A

$$\hat{U}_{BS}(\eta)|1,0\rangle_{AB} = \hat{U}_{BS}(\eta)\hat{a}^{\dagger}|vac\rangle = \hat{U}_{BS}(\eta)\hat{a}^{\dagger}\hat{U}^{\dagger}_{BS}(\eta)\hat{U}_{BS}(\eta)|vac\rangle \tag{3.137}$$

$$= (\sqrt{\eta}\,\hat{a}^{\dagger} + \sqrt{1-\eta}\,\hat{b}^{\dagger})|vac\rangle \tag{3.138}$$

$$= \sqrt{\eta}\,|1,0\rangle_{AB} + \sqrt{1-\eta}\,|0,1\rangle_{AB}. \tag{3.139}$$

Similarly,

$$\hat{U}_{BS}(\eta)|0,1\rangle_{AB} = \sqrt{1-\eta}\,|1,0\rangle_{AB} - \sqrt{\eta}\,|0,1\rangle_{AB}. \tag{3.140}$$

So a single photon in one mode is transformed into a superposition of photons in each mode. The states (3.139) and (3.140) are entangled for any $0 < \eta < 1$. In general, for most input states, a beam splitter will generate correlations between the modes. Coherent states, however, are an important exception to this rule. When the inputs are both coherent states, the transformation is simply

$$\hat{U}_{BS}(\eta)|\alpha,\beta\rangle_{AB} = |\sqrt{\eta}\alpha + \sqrt{1-\eta}\beta, \sqrt{1-\eta}\,\alpha - \sqrt{\eta}\,\beta\rangle_{AB}, \tag{3.141}$$

where the output is again a product of coherent states. In particular, for a balanced beam splitter, $\eta = \frac{1}{2}$, we have

$$\hat{U}_{BS}|\alpha,\beta\rangle_{AB} = |\frac{\alpha+\beta}{\sqrt{2}}, \frac{\alpha-\beta}{\sqrt{2}}\rangle_{AB}. \tag{3.142}$$

This also illustrates interference at the beam splitter. If there is only light in one of the input modes (i.e. α or β is zero), then it will be distribued evenly into the two output modes. However, if the input amplitudes are equal, $\beta = \alpha$, then there will be complete destructive interference in mode B and all the light will exit in A. For states that are 'less classical' than coherent states, the interference can have interesting effects, such as bunching (see Example 3.2)

[6]Note that different phase conventions for the beam-splitter transformation exist in the literature. For example [5] assumes that reflections cause a $\pi/2$ phase shift.

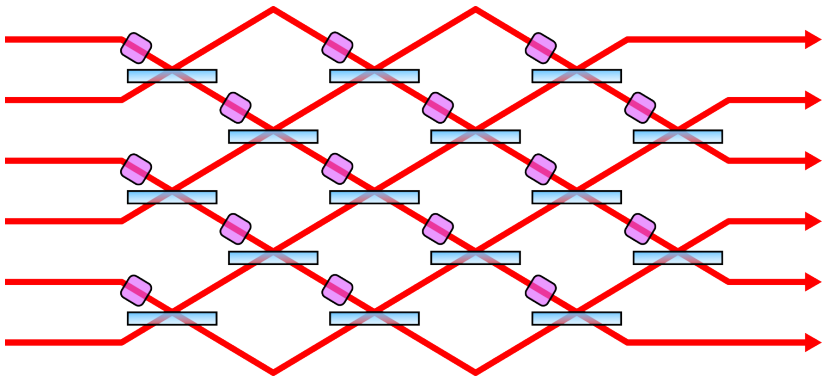

Figure 3.12: Beam-splitter network for realising any passive linear-optical transformation on $N = 6$ modes. A total of 15 beam splitters (blue) and phase shifters (magenta) are needed. Other elements needed to direct the light (e.g. mirrors) are not shown for simplicity. The generalisation to more modes is straightforward.

Example 3.2: Photon bunching

When photons interfere at a beam splitter, they tend to bunch into states with multiple photons occupying the same modes. Fundamentally, this is a consequence of photons being bosons and obeying the canonical commutation relations (3.19) (fermions such as electrons, on the other hand, would anti-bunch in accordance with the Pauli exclusion principle).

Consider two single photons incident on a balanced beam splitter. The state transforms as

$$\hat{U}_{BS}|1,1\rangle_{AB} = \frac{1}{2}(\hat{a}^\dagger + \hat{b}^\dagger)(\hat{a}^\dagger - \hat{b}^\dagger)|vac\rangle \tag{3.143}$$

$$= \frac{1}{2}((\hat{a}^\dagger)^2 - (\hat{b}^\dagger)^2)|vac\rangle \tag{3.144}$$

$$= \frac{1}{\sqrt{2}}(|2,0\rangle_{AB} - |0,2\rangle_{AB}). \tag{3.145}$$

That is, the two photons always exit in the same output mode, while one photon at each output never occurs.

For free-space optics, beam splitters are usually implemented using cubes of glass consisting of two triangular prisms glued together, with the splitting happening at the interface between them, or by 'half-silvered mirrors' made of glass or plastic with a thin metallic coating applied. In optical fibre, beam splitters can be implemented by bringing two fibre cores sufficiently close to each other (or coupling into wave guides that are brought close), allowing for evanescent coupling. Fig. 3.11 shows several standard ways of drawing beam splitters often seen in scientific papers.

Note that only modes with identical properties will interfere at a physical beam splitter. That is, the frequencies, polarisations, spatial profiles, timing